

ENME608 Mechanisms

Exam Preparation — Six Question Types Worked in Detail

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Q1 — SVAJ (Cam Motion, work in RADIANS)

Exam focus. Use radians throughout. The segment angle β and the cam angle θ must both be in radians, otherwise the v, a, j magnitudes are wrong by factors of $(180/\pi)$. Derivatives here are with respect to θ (not time).

Definitions and $\theta \rightarrow$ time conversion

$$s = s(\theta) \text{ [length]}, \quad v = \frac{ds}{d\theta}, \quad a = \frac{d^2s}{d\theta^2}, \quad j = \frac{d^3s}{d\theta^3}.$$

For constant cam speed ω (rad/s), convert the kinematic (θ -based) values to *time* derivatives:

$$\dot{s} = v\omega, \quad \ddot{s} = a\omega^2, \quad \dddot{s} = j\omega^3.$$

Standard double-dwell rise functions

Cycloidal:

$$s = h \left[\frac{\theta}{\beta} - \frac{1}{2\pi} \sin \frac{2\pi\theta}{\beta} \right], \quad v = \frac{h}{\beta} \left[1 - \cos \frac{2\pi\theta}{\beta} \right],$$
$$a = \frac{2\pi h}{\beta^2} \sin \frac{2\pi\theta}{\beta}, \quad j = \frac{4\pi^2 h}{\beta^3} \cos \frac{2\pi\theta}{\beta}.$$

3-4-5 polynomial (with $u = \theta/\beta$):

$$s = h [10u^3 - 15u^4 + 6u^5], \quad v = \frac{h}{\beta} [30u^2 - 60u^3 + 30u^4],$$
$$a = \frac{h}{\beta^2} [60u - 180u^2 + 120u^3], \quad j = \frac{h}{\beta^3} [60 - 360u + 360u^2].$$

Both give $v = a = 0$ at the ends ($u = 0, 1$), so they match a dwell on each side (continuous s, v, a ; finite j — satisfies the *fundamental law of cam design*).

Worked example (cycloidal rise)

Given:

$h = 1$ in rise, segment $\beta = \pi/2$ rad (90°), cam speed $\omega = 10$ rad/s. Evaluate at $\theta = \beta/4 = \pi/8$ rad.

Solution:

Phase angle $\frac{2\pi\theta}{\beta} = 2\pi \cdot \frac{\pi/8}{\pi/2} = \frac{\pi}{2}$.

$$s = 1 \left[\frac{1}{4} - \frac{1}{2\pi} \sin \frac{\pi}{2} \right] = 0.250 - 0.159 = \boxed{0.091 \text{ in}}$$

$$v = \frac{1}{\pi/2} \left[1 - \cos \frac{\pi}{2} \right] = \frac{2}{\pi} (1 - 0) = \boxed{0.637 \text{ in/rad}}$$

$$a = \frac{2\pi(1)}{(\pi/2)^2} \sin \frac{\pi}{2} = \frac{8}{\pi} = \boxed{2.546 \text{ in/rad}^2}, \quad j = \frac{4\pi^2(1)}{(\pi/2)^3} \cos \frac{\pi}{2} = \boxed{0}$$

Time derivatives at this instant:

$$\dot{s} = v\omega = 6.37 \text{ in/s}, \quad \ddot{s} = a\omega^2 = 254.6 \text{ in/s}^2, \quad \dddot{s} = j\omega^3 = 0.$$

Q2 — Gear Train: Shaft Speeds and Reflected Inertia

(a,b) Speed of a shaft / output ω

For each mesh the velocity ratio is

$$m_V = \frac{\omega_{\text{out}}}{\omega_{\text{in}}} = \mp \frac{N_{\text{in}}}{N_{\text{out}}} \quad (\text{external } -, \text{ internal } +).$$

Compound train (gears on a common shaft share speed):

$$m_V = \pm \frac{\prod N_{\text{driving}}}{\prod N_{\text{driven}}}, \quad \omega_{\text{out}} = m_V \omega_{\text{in}}.$$

Planetary / epicyclic — two-step (relative-to-arm) method:

$$\frac{\omega_L - \omega_{\text{arm}}}{\omega_F - \omega_{\text{arm}}} = \pm \frac{N_F}{N_L},$$

where F = first gear, L = last gear of the chain taken *relative to the arm*.

(c) Effective (reflected) inertia

Inertia reflected to a reference shaft scales with the *square* of the speed ratio:

$$I_{\text{eff}} = I_{\text{load}} \left(\frac{\omega_{\text{load}}}{\omega_{\text{ref}}} \right)^2 = \frac{I_{\text{load}}}{m_V^2} \quad (\text{reflecting a load back to the motor}).$$

Total inertia felt at the motor = motor rotor inertia + $\sum$ reflected inertias of all downstream rotating parts.

Worked example

Given:

Motor at $\omega_{\text{in}} = 1500$ rpm drives a compound train: mesh 1 $N_A = 20 \rightarrow N_B = 60$ (external), shaft B also carries $N_C = 18 \rightarrow N_D = 54$ (external). Output load inertia $I_L = 0.5 \text{ blob} \cdot \text{in}^2$ on the output shaft.

Find:

output speed and the inertia reflected to the motor shaft.

Solution:

$$m_V = \left(-\frac{20}{60} \right) \left(-\frac{18}{54} \right) = (+) \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{9},$$

$$\omega_{\text{out}} = \frac{1}{9}(1500) = \boxed{166.7 \text{ rpm}} \quad (\text{same sense as motor, } (-1)^2 = +).$$

Reflected inertia at the motor:

$$I_{\text{eff}} = I_L \left(\frac{\omega_{\text{out}}}{\omega_{\text{in}}} \right)^2 = 0.5 (1/9)^2 = 0.5/81 = \boxed{6.17 \times 10^{-3} \text{ blob} \cdot \text{in}^2}.$$

Exam focus. Reflecting *toward the slower shaft* multiplies inertia; reflecting toward the *faster* (motor) shaft divides it by m_V^2 . Always square the ratio.

Q3 — Gears: Contact Length, Contact Ratio, Torque

(a) Length of action Z (contact length)

$$Z = \sqrt{(r_p + a_p)^2 - (r_p \cos \phi)^2} + \sqrt{(r_g + a_g)^2 - (r_g \cos \phi)^2} - C \sin \phi,$$

where r_p, r_g are pitch radii, $a_p = a_g = 1/P_d$ the addenda, ϕ the pressure angle, and $C = r_p + r_g$ the centre distance.

(b) Contact ratio m_p

$$m_p = \frac{Z}{p_b} = \frac{Z P_d}{\pi \cos \phi}, \quad p_b = p_c \cos \phi = \frac{\pi}{P_d} \cos \phi.$$

Require $m_p \geq 1.2$ (average teeth in contact). $m_p \geq 1$ is the bare minimum for continuous transmission.

(c) Torque transmitted by a gear

Power is constant through a mesh, so

$$T_{\text{out}} = T_{\text{in}} \frac{\omega_{\text{in}}}{\omega_{\text{out}}} = T_{\text{in}} m_T, \quad m_T = \frac{1}{m_V}.$$

The tangential force at the pitch circle and the torque are linked by

$$W_t = \frac{T}{r}, \quad T = W_t r.$$

Worked example

Given:

Pinion $N_p = 19$, gear $N_g = 37$, $P_d = 6 \text{ in}^{-1}$, $\phi = 20^\circ$, full-depth ($a = 1/P_d = 0.1667 \text{ in}$). Pinion input torque $T_p = 100 \text{ lb}\cdot\text{in}$.

Solution:

Pitch radii: $r_p = N_p/(2P_d) = 19/12 = 1.583$, $r_g = 37/12 = 3.083 \text{ in}$; $C = 4.667 \text{ in}$.

$$\begin{aligned} Z &= \sqrt{(1.583+0.167)^2 - (1.583 \cos 20^\circ)^2} + \sqrt{(3.083+0.167)^2 - (3.083 \cos 20^\circ)^2} - 4.667 \sin 20^\circ \\ &= \sqrt{3.063 - 2.213} + \sqrt{10.563 - 8.395} - 1.596 = 0.922 + 1.472 - 1.596 = \boxed{0.798 \text{ in}}. \end{aligned}$$

$$m_p = \frac{Z P_d}{\pi \cos \phi} = \frac{0.798(6)}{\pi \cos 20^\circ} = \frac{4.788}{2.952} = \boxed{1.62} \quad (\geq 1.2 \checkmark)$$

Torque on the gear (external mesh, $m_T = N_g/N_p$):

$$T_g = T_p \frac{N_g}{N_p} = 100 \cdot \frac{37}{19} = \boxed{194.7 \text{ lb}\cdot\text{in}}.$$

Q4 — FBD of a Link in a Four-Bar Linkage

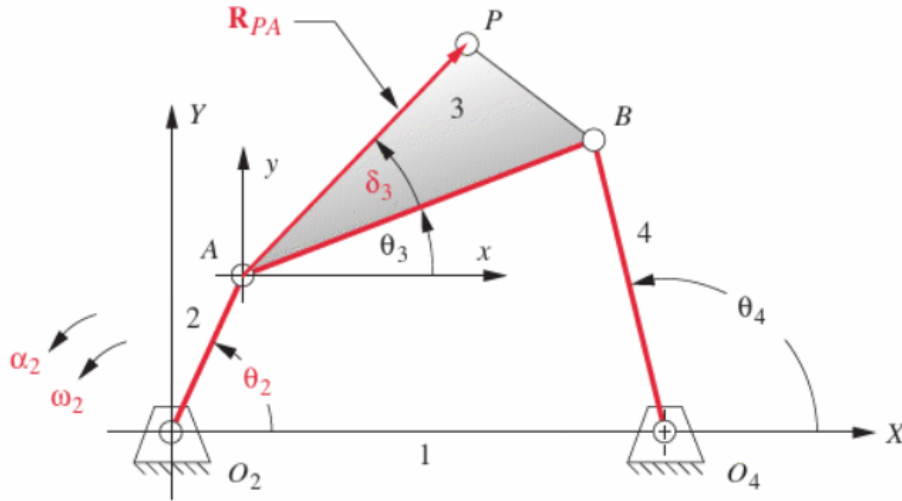


FIGURE P7-1

Four-bar reference (Figure P7-1): links 2 (crank), 3 (coupler), 4 (rocker).

(a) Force and torque equations per link

For *each* link write the planar equations of motion (Newton–Euler):

$$\sum F_x = m a_{Gx}, \quad \sum F_y = m a_{Gy}, \quad \sum T_G = I_G \alpha.$$

Notation: $\mathbf{F}_{ij}$ = force on link j by link i , with $\mathbf{F}_{ji} = -\mathbf{F}_{ij}$. $\mathbf{R}_{ij}$ = vector from the link's CG to the joint. Scalar torque of a force: $T = R_x F_y - R_y F_x$.

Link 2 (crank), pinned to ground at O_2 , joined to link 3 at A , driving torque T_{12} :

$$F_{12x} + F_{32x} = m_2 a_{G2x}, \quad F_{12y} + F_{32y} = m_2 a_{G2y},$$

$$T_{12} + (R_{12x} F_{12y} - R_{12y} F_{12x}) + (R_{32x} F_{32y} - R_{32y} F_{32x}) = I_{G2} \alpha_2.$$

Link 3 (coupler), joints at A and B , external force $\mathbf{F}_P$ at $\mathbf{R}_P$:

$$F_{23x} + F_{43x} + F_{Px} = m_3 a_{G3x}, \quad F_{23y} + F_{43y} + F_{Py} = m_3 a_{G3y},$$

$$(R_{23x} F_{23y} - R_{23y} F_{23x}) + (R_{43x} F_{43y} - R_{43y} F_{43x}) + (R_{Px} F_{Py} - R_{Py} F_{Px}) = I_{G3} \alpha_3.$$

Link 4 (rocker), joints at B and ground O_4 :

$$F_{34x} + F_{14x} = m_4 a_{G4x}, \quad F_{34y} + F_{14y} = m_4 a_{G4y},$$

$$(R_{34x} F_{34y} - R_{34y} F_{34x}) + (R_{14x} F_{14y} - R_{14y} F_{14x}) = I_{G4} \alpha_4.$$

These nine scalar equations (with $\mathbf{F}_{23} = -\mathbf{F}_{32}$, $\mathbf{F}_{34} = -\mathbf{F}_{43}$) solve for the nine unknowns F_{12x} , F_{12y} , F_{32x} , F_{32y} , F_{43x} , F_{43y} , F_{14x} , F_{14y} , T_{12} .

(b) Equation for the crank-driving torque

Isolating T_{12} from link 2's moment equation:

$$T_{12} = I_{G2} \alpha_2 - (R_{12x} F_{12y} - R_{12y} F_{12x}) - (R_{32x} F_{32y} - R_{32y} F_{32x}).$$

Equivalently, the whole system is the scalar **virtual-work** statement (fast route to T_{12} alone, no joint forces needed):

$$T_{12} \omega_2 = \sum_i (m_i \mathbf{a}_{Gi} \cdot \mathbf{V}_{Gi} + I_{Gi} \alpha_i \omega_i) - \sum \mathbf{F}_P \cdot \mathbf{V}_P - \sum T_k \omega_k.$$

Exam focus. If the exam only asks for the *driving torque*, virtual work is much quicker than inverting the 9×9 . If it asks for *joint reaction forces*, you must use the matrix (Newtonian) method.

Q5 — Crank-Slider: Open and Crossed Position

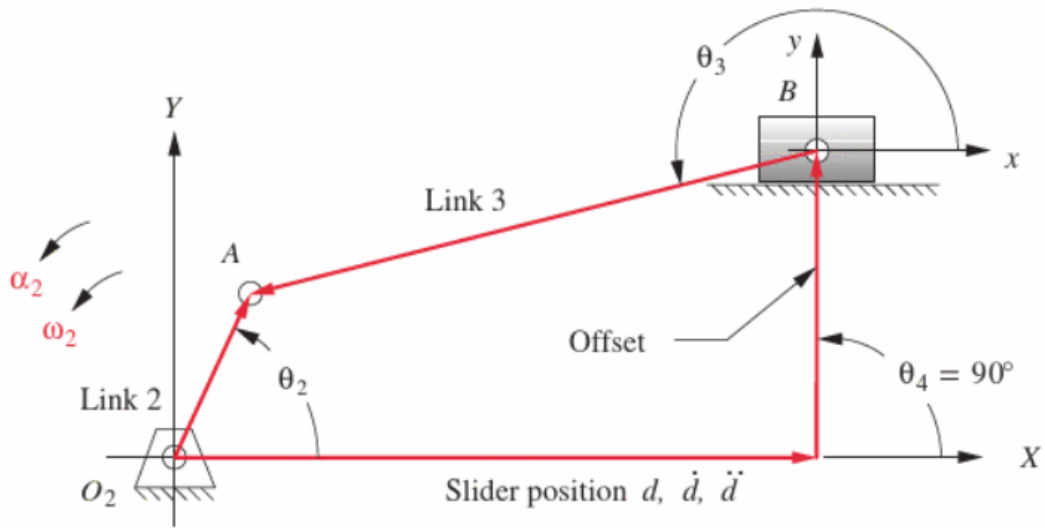


FIGURE P7-2

Offset crank-slider (Figure P7-2): crank a , coupler b , offset, slider d .

(a,b) The two configurations and θ_3

The offset crank-slider loop (a crank, b coupler, c offset) gives two assembly modes:

$$\sin \theta_3 = \frac{a \sin \theta_2 - c}{b} \Rightarrow \theta_{3,\text{open}} = \arcsin \frac{a \sin \theta_2 - c}{b}, \quad \theta_{3,\text{crossed}} = 180^\circ - \theta_{3,\text{open}}.$$

(d) Slider position d

$$d = a \cos \theta_2 - b \cos \theta_3.$$

Evaluate with each θ_3 to get d_{open} and d_{crossed} .

Worked example (Row a)

Given:

$a = 1.4$, $b = 4$, offset $c = 1$, $\theta_2 = 45^\circ$.

Solution:

$$\sin \theta_3 = \frac{1.4 \sin 45^\circ - 1}{4} = \frac{0.990 - 1}{4} = -0.00250.$$

$$\theta_{3,\text{open}} = \arcsin(-0.00250) = \boxed{-0.14^\circ} (\equiv 359.86^\circ), \quad \theta_{3,\text{crossed}} = 180^\circ + 0.14^\circ = \boxed{180.14^\circ}.$$

$$d_{\text{open}} = 1.4 \cos 45^\circ - 4 \cos(-0.14^\circ) = 0.990 - 4.000 = \boxed{-3.01} \text{ (slider left of } O_2),$$

$$d_{\text{crossed}} = 1.4 \cos 45^\circ - 4 \cos(180.14^\circ) = 0.990 + 4.000 = \boxed{4.99} \text{ (slider right of } O_2).$$

Exam focus. The *physical* assembly that matches Figure P7-2 (slider to the right) is the $d = 4.99$, $\theta_3 \approx 0.14^\circ$ branch. Labelling of “open/crossed” depends on the figure’s chosen loop direction — always state which root you keep and sketch it.

Q6 — Crank-Slider: Velocity and Acceleration

Using the physically-assembled branch ($\theta_3 = 0.14^\circ$, $d = 4.99$), with $\omega_2 = 10 \text{ rad/s}$ and $\alpha_2 = 0$.

(a) Velocity of point A — magnitude and direction

A is the crank pin, pure rotation about O_2 :

$$|\mathbf{V}_A| = a \omega_2 = (1.4)(10) = 14 \text{ in/s},$$

direction $\perp$ to the crank O_2A , i.e. $\theta_2 + 90^\circ = 135^\circ$:

$$\mathbf{V}_A = a \omega_2 (-\sin \theta_2, \cos \theta_2) = (-9.90, 9.90) \text{ in/s} \Rightarrow \boxed{14 \text{ in/s at } 135^\circ}.$$

(b) Slip velocity of the slider

First the coupler angular velocity, then the slider speed:

$$\omega_3 = -\frac{a \omega_2 \cos \theta_2}{b \cos \theta_3} = -\frac{(1.4)(10) \cos 45^\circ}{4 \cos 0.14^\circ} = -2.48 \text{ rad/s},$$

$$\dot{d} = -a \omega_2 \sin \theta_2 - b \omega_3 \sin \theta_3 = -9.90 - (4)(-2.48)(0.0025) = \boxed{-9.88 \text{ in/s}} \text{ (slip, in } -X).$$

(c) Acceleration of points A and B

$$\alpha_3 = \frac{a \omega_2^2 \sin \theta_2 + b \omega_3^2 \sin \theta_3}{b \cos \theta_3} = \frac{(1.4)(100) \sin 45^\circ + (4)(2.48^2)(0.0025)}{4 \cos 0.14^\circ} = 24.8 \text{ rad/s}^2.$$

$$\mathbf{a}_A = -a \omega_2^2 (\cos \theta_2, \sin \theta_2) = (-99.0, -99.0) \text{ in/s}^2, \quad \boxed{|\mathbf{a}_A| = 140 \text{ in/s}^2}.$$

$$\ddot{d} = -a \omega_2^2 \cos \theta_2 - b \alpha_3 \sin \theta_3 - b \omega_3^2 \cos \theta_3 = -99.0 - 0.25 - 24.6 = \boxed{-123.7 \text{ in/s}^2} (= \mathbf{a}_B, \text{ along slide}).$$

(d) Tangential and centripetal components

Point A (on crank, radius a):

$$a_A^t = a \alpha_2 = 0, \quad a_A^n = a \omega_2^2 = (1.4)(100) = \boxed{140 \text{ in/s}^2} \text{ (toward } O_2).$$

Since $\alpha_2 = 0$, $\mathbf{a}_A$ is purely centripetal.

Point B relative to A (coupler, length b):

$$a_{BA}^t = b \alpha_3 = (4)(24.8) = \boxed{99.2 \text{ in/s}^2} (\perp \text{ coupler}),$$

$$a_{BA}^n = b\omega_3^2 = (4)(2.48)^2 = \boxed{24.6 \text{ in/s}^2} \text{ (along coupler, } B \rightarrow A\text{)}.$$

The slider B itself translates along the slide, so its absolute acceleration has no centripetal part — it is the single linear value $\ddot{d} = -123.7 \text{ in/s}^2$; the tangential/centripetal split above describes the coupler motion that produces it.

Exam focus. No Coriolis term appears here: the slide axis is fixed (does not rotate). Coriolis ($2\omega v_{\text{rel}}$) only enters when a block slides along a *rotating* link (inverted slider-crank).